Exercise: 4

Mitosis in onion root tip

Aim: To study and identify the mitosis stages – Metaphase and Anaphase.

Principle: Somatic growth of both plants and animals takes place by increase in the number of cells. The cells divide mitotically wherein number of chromosomes remains unchanged in the daughter cells from that in the maternal cells. Cells from the growing root-tips and apex of shoot buds are suitable for mitotically dividing cells. In animals mitotically dividing cells can be easily scored from the bone marrow of a vertebrate. The cell from the epithelium of gills in fishes and from the tail of growing tadpole larvae of frog are also good sources for scoring the mitotically dividing cells.

Requirements: Onion root, HCl, Safranin stain, slide, Coverslip, Permanent slides Compound microscope.



1. Cut the tip 5 to 8 mm from the tip of the freshly sprouted onion root. Discard the rest of the root.
2. Wash them in water on a clean microscope slide.
3. Place one drop of 1N HCl on the root tip and add 2-3 drops of Safranin/ Acetocarmine stain to the slide.
4. Warm the slide gently over the alcohol lamp for about one minute. (Do not allow the slide to get hot to the touch).
5. Carefully blot the excess stain with a blotting paper.
6. After (10 to 20 seconds) put one or two drops of water and blot them carefully using blotting paper.
7. Again put a drop of water on the root tip and mount a cover slip on it avoiding air bubbles.
8. Squash the slide with your thumb using a firm and even pressure. (Avoid squashing with such force that the cover slip breaks or slides).
9. Observe it under a compound microscope in 10x objective. Scan and narrow down to a region containing dividing cells and switch to 40x for a better view.

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